GHENA HANI

AI Engineer | Deep Learning · NLP · LLM Systems

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github.com/HANIDEVINF | linkedin.com/in/ghena-hani-3543a3375 | Portfolio

PROFESSIONAL SUMMARY

AI engineer and Master's student in Artificial Intelligence (USTHB) with five AI research internships at CERIST and about two years of freelance full-stack delivery. Builds deep learning systems end to end: a multilingual Transformer architecture for Arabic dialects, Berber languages, French and English (MuCAT), a leakage-free ECG arrhythmia classifier compressed to a 155 KB INT8 TFLite model, and a real-time IoT patient-monitoring platform connected to physical sensors.

TECHNICAL SKILLS

- **Deep Learning & ML:** PyTorch, TensorFlow/Keras, Hugging Face Transformers, scikit-learn, BERT-family models, NLP, time-series and signal processing, TFLite INT8 quantization
- **LLM Applications:** RAG and ReAct/tool-calling prototypes, LLM evaluation harnesses, guardrails, prompt-injection awareness
- **Engineering:** Python, FastAPI, Flask, TypeScript, Next.js/React, REST, WebSockets/WebRTC, Docker, Git, MongoDB, PostgreSQL, Supabase, Flutter, Raspberry Pi

EXPERIENCE

AI Research Intern (5 internships) 2024 – Present

CERIST (Research Center on Scientific and Technical Information), Algiers

- Designed and implemented **MuCAT**, a multilingual NLP architecture (Arabic, Algerian Darija, Kabyle, Chaoui, French, English) on mDeBERTa-v3-base with hierarchical attention pooling, FiLM language-aware gating and an evidential (Dirichlet) uncertainty head.
- Built an ECG arrhythmia classifier (1D-CNN + 2 Transformer blocks + 8 RR-interval features) and deployed it as an INT8 TFLite model for embedded inference (see Projects).
- Developed real-time ECG and blood-glucose anomaly detection connected to physical medical sensors, with emergency alert workflows.
- Contributing to AAPIS (Algerian Artificial Pancreas Intelligent System): embedded AI, IoT sensor fusion, biosignals.

Independent Full-Stack & AI Developer ∼2 years

Freelance, Algiers

- Delivered production systems for Algerian clients: a web platform with doctor and receptionist Electron desktop apps for Tadjmeel Clinica (Algiers); a multi-role ERP/POS with a G50 fiscal ledger and 58-wilaya delivery tracking; e-commerce and inventory systems for menswear boutiques in Algiers, Sidi Bel Abbès and Mascara.
- Built and deployed 7 public interactive AI demos and prototypes (speech, content moderation, document classification, agent workflows, LLM evaluation) with source on GitHub.

SELECTED PROJECTS

ECG Arrhythmia Classification (v7) GitHub

- 1D-CNN + 2 Transformer blocks + 8 RR features on MIT-BIH with a strict inter-patient DS1/DS2 split: 87.63% single model and 90.13% 5-fold ensemble accuracy (0.529 macro-F1) on 49,668 unseen beats.
- INT8 TFLite model of 155.4 KB, 0.32 ms per beat, 99.82% agreement with the float model.

MedGuardAI – Real-Time Patient Monitoring (Licence final-year project, USTHB) GitHub

- Platform linking physical ECG and glucose sensors (Raspberry Pi) to AI anomaly detection, emergency alerts, patient–doctor chat, WebRTC consultations, appointments and generated prescriptions. Flutter, Flask, MongoDB.

MCTNet Crop-Type Classification (Sentinel-2) – Neural Networks project, USTHB GitHub

- Reproduced a CNN–Transformer crop-mapping network in Keras on Sentinel-2 time series I extracted with Google Earth Engine (Arkansas, California); fixed a normalization leakage bug and ablated climate, soil and topography covariates.
- Extended it with a BiGRU and vegetation indices: 0.966 OA (Arkansas) and 0.846 OA (California) vs 0.937 and 0.791 for the baseline reproduction. Single runs; gains confounded by a larger training set.

EDUCATION

Master in Artificial Intelligence Sep 2025 – Expected June 2027

University of Science and Technology Houari Boumediene (USTHB), Algiers

Licence in Computer Science Sep 2022 – June 2025

USTHB, Algiers – final-year project: MedGuardAI

Certification: Deep Learning (Code 213), final project completed, Nov 2025 – Jun 2026

Languages: Arabic (native) | French (full professional) | English (professional working)